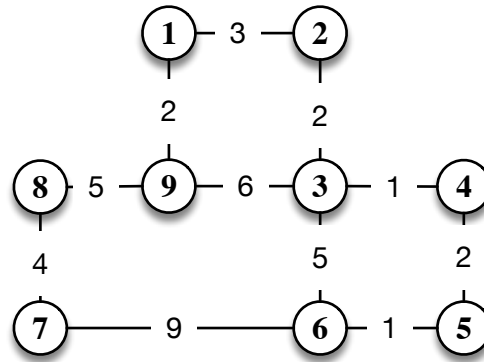


Second part: Open book

Total score: 10 pts.

You are warmly invited to be tidy and concise. Messy papers will be penalized.

1. Consider the undirected graph in the figure below, where lengths are reported on the arcs. Is this graph Eulerian? Why? In case it is not, it can be made Eulerian by adding suitable arcs, corresponding to paths in the original graph. Find a minimum length extension of the graph that makes it Eulerian. Justify the answer.



2. Consider the following Linear Programming problem resulting from a production planning:

$$\begin{aligned}
 \max \quad & 5x_1 + 5x_2 \\
 A) \quad & -x_1 + 3x_2 \leq 12 \\
 B) \quad & -2x_1 + x_2 \leq 2 \\
 C) \quad & 2x_1 - x_2 \leq 6 \\
 & x_1, x_2 \geq 0
 \end{aligned}$$

where variables x_1 and x_2 represent the quantity of product 1 and 2 respectively, and A), B) and C) represent the resource constraints. Prove that the solution $x = \begin{bmatrix} 6 \\ 6 \end{bmatrix}$ is the optimal one.

Consider now the introduction of three new products 3, 4, and 5, corresponding to variables x_3, x_4 and x_5 , whose coefficients in the objective function and in the constraints are summarized in the following table:

prod.	3	4	5
o.f.	7	13	10
A)	1	2	3
B)	0	2	1
C)	1	2	0

What are the products whose addition to the original problem can increase the value of the optimal solution? Justify the answer.